4. Algorithms on graphs

4.1 Shortest path problems

Definition 4.1 Let $G = (V, E)$ be a directed graph. A **weight function**, sometimes also called **cost function**, for the edges of G is a mapping $w : E \to \mathbb{R}$. If $\pi = v_0, v_1, \ldots, v_r$ is a path in G, then the value

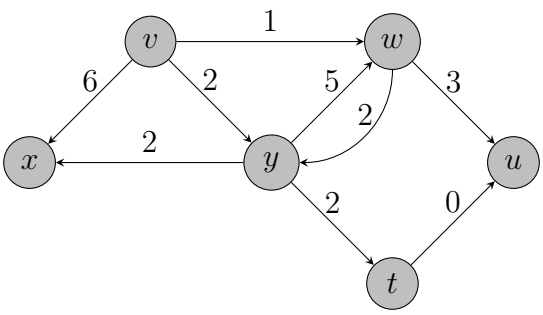
$$\omega(\pi) = \sum_{i=0}^{r-1} \omega(v_i, v_{i+1})$$

is its **length** with respect to ω. The triple $G = (V, E, \omega)$ is called **weighted graph**.

Definition 4.2 Let $G = (V, E, \omega)$ be a weighted graph and $v, w \in V$. A **shortest path** from v to w in G with respect to ω is an v-w-path π with $\omega(\pi) \leq \omega(\pi')$ for any other v-w-path π'. The **length** $\delta(v, w)$ of the shortest path from v to w is defined by

$$\delta(v, w) := \begin{cases} \min\{\omega(\pi) : \pi \text{ is a path from } v \text{ to } w\}, & \text{if such a path exists,} \\ \infty, & \text{otherwise.} \end{cases}$$

Example 4.3 Consider the weighted graph $G = (V, E, \omega)$ given by



A shortest path from v to x is $\pi = v, y, x$. Its length is $\omega(\pi) = 2 + 2 = 4$ and is shorter than $\omega(x, y) = 6$. There are two shortest paths from v to the vertex u, namely $\pi_1 = v, w, u$ and $\pi_2 = v, y, t, u$ with $\omega(\pi_1) = \omega(\pi_2) = \delta(v, u) = 4$. △

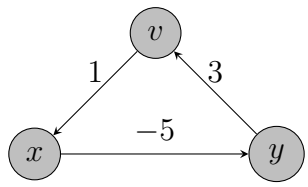
One distinguishes different variants of *shortest path problems* for a given weighted graph $G = (V, E, \omega)$.

1. **Single pair shortest path problem:** a shortest path between a given pair of vertices $v, w \in V$ is sought.
2. **Single source shortest path problem:** shortest paths from a vertex $v \in V$ to all $w \in \mathrm{post}^\star(v)$ are sought.
3. **All pair shortest path problem:**
   shortest paths between any pair of vertices $v, w \in V$ are sought.

The first problem can of course be solved by the second one. Indeed, no asymptotically cheaper method is not known. For this reason, we will consider this more general problem.

Negative weights are not excluded according to Definition 4.1. Nonetheless, cycles of negative length are problematic as the subsequent example illustrates:

Example 4.4 Consider the following weighted graph $G = (V, E, \omega)$:



There is no shortest path in this graph, since

$$\begin{aligned} \omega(v, x, y) &= -4, \\ \omega(v, x, y, v, x, y) &= -5, \\ \omega(v, x, y, v, x, y, v, x, y) &= -6, \\ &\vdots \end{aligned}$$

△

Lemma 4.5 Let $G = (V, E, \omega)$ be a weighted graph. If there are no cycles of negative length in G, then there exists a shortest path π in G for each pair of vertices $v, w \in V$ with $w \in \mathrm{post}^\star(v)$ satisfying

$$\delta(v, w) = \omega(\pi) > -\infty.$$

Proof. As there a no cycles of negative length, it suffices to consider all simple paths from v to w. These are only finitely many since $|V|$ and $|E|$ are finite, which implies the assertion. □

Lemma 4.6 Let $G = (V, E, \omega)$ be a weighted graph without cycles of negative length. If $\pi = v_0, v_1, \ldots, v_{r-1}, v_r$ is a shortest path from v_0 to v_r, then the partial paths $\pi_{i,j} = v_i, v_{i+1}, \ldots, v_j$ of π are shortest paths from v_i to v_j for all $0 \le i < j \le r$.

Proof. Assume that there exists a shortest path $\pi'_{i,j}$ from v_i to v_j with $\omega(\pi'_{i,j}) < \omega(\pi_{i,j})$. Then, the composed path $\widehat{\pi} = \pi_{1,i}, \pi'_{i,j}, \pi_{j,r}$ satisfies the estimate

$$\omega(\widehat{\pi}) = \pi_{1,i} + \pi'_{i,j} + \pi_{j,r} < \pi_{1,i} + \pi_{i,j} + \pi_{j,r} = \omega(\pi).$$

This is a contradiction to the assumption that π is a shortest path from v_0 to v_r. $\square$

Corollary 4.7 Let $G = (V, E, \omega)$ be a weighted graph without cycles of negative length and $\pi = v_0, v_1, \ldots, v_r$ a shortest path from v_0 to v_r. Then, it holds

$$\delta(v_0, v_r) = \delta(v_0, v_{r-1}) + \omega(v_{r-1}, v_r).$$

Proof. According to Lemma 4.6, the path $\pi' = v_0, v_1, \ldots, v_{r-1}$ is a shortest path from v_0 to v_{r-1}, i.e., $\omega(\pi') = \delta(v_0, v_{r-1})$. This means

$$\delta(v_0, v_r) = \omega(\pi) = \omega(\pi') + \omega(v_{r-1}, v_r).$$

$\square$

Based on this corollary, the following algorithm solves the single source shortest path problem in the case on *nonnegative* weights:

Algorithm 4.8 (Dijkstra)

input: A weighted graph $G = (V, E, \omega)$ with nonnegative weights and initial vertex $s \in V$.

output: Shorted paths from s to all $v \in V$ together with the length $\ell(v)$. Precisely, $\ell(v)$ is the length of the shortest s-v-path, which consists of the shortest s-$p(v)$-path and the edge $\big(p(v), v\big)$. For $v \notin \mathrm{post}^\star(s)$, $\ell(v) = \infty$ and $p(v)$ is undefined.

① set $\ell(s) := 0$ and $\ell(v) := \infty$ for all $v \in V \setminus \{s\}$, set $R := \emptyset$

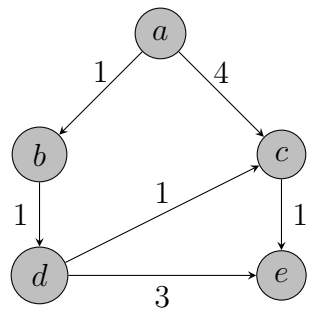
② find $u \in V \setminus R$ with $\ell(u) = \min_{v \in V \setminus R} \ell(v)$

③ set $R := R \cup \{u\}$

④ for all $v \in V \setminus R$ with $(u, v) \in E$:
  if $\ell(v) > \ell(u) + \omega(u, v)$, then
    set $\ell(v) := \ell(u) + \omega(u, v)$ and $p(v) := u$

⑤ if $R \neq V$, then go to ②

Example 4.9 Consider the weighted graph $G = (V, E, \omega)$ given by



Starting from the initial vertex a, the algorithm performs the following steps:

Iteration	a	b	c	d	e	u	$\ell(u)$	$p(u)$
0	0	∞	∞	∞	∞	a	0	—
1	—	1	4	∞	∞	b	1	a
2	—	—	4	2	∞	d	2	b
3	—	—	3	—	5	c	3	d
4	—	—	—	—	4	e	4	c

△

Theorem 4.10 (Dijkstra) Dijkstra's algorithm is working correctly. Its runtime is $\mathcal{O}(n^2)$ with $n = |V|$.

Proof. For the sake of clarity, we add the iteration index as a suffix to all variables of Dijkstra's algorithm. We proof that the following propositions are true at each execution of ②:

(a) For all $v \in R^{(k)}$ and all $y \in V \setminus R^{(k)}$, there holds $\ell^{(k)}(v) \leq \ell^{(k)}(y)$.

(b) For all $v \in R^{(k)}$, $\ell^{(k)}(v)$ is the length of the shortest path from s to v. If $\ell^{(k)}(v) < \infty$ and $v \neq s$, then there exists a shortest path from s to v with vertices from $R^{(k)}$ and last edge $\big(p^{(k)}(v), v\big)$.

(c) For all $v \in V \setminus R^{(k)}$, $\ell^{(k)}(v)$ is the length of the shortest path from s to v in that subgraph of G which consists of the vertices $R^{(k)} \cup \{v\}$. If $\ell^{(k)}(v) \neq \infty$, then $p^{(k)}(v) \in R^{(k)}$ and

$$\ell^{(k)}(v) = \ell^{(k)}\big(p^{(k)}(v)\big) + w\big(p^{(k)}(v), v\big).$$

As these proposition are true for $k = 0$, i.e., after the execution of ①, we show that these propositions remain true after the execution of ③ and ④, i.e., we show the induction step $k \mapsto k + 1$. To this end, let $k \geq 0$ be arbitrary and u the vertex chosen in ②.

Due to (a), it holds for arbitrary $v \in R^{(k)}$ and $y \in V \setminus R^{(k)}$ that

$$\ell^{(k)}(v) \leq \ell^{(k)}(u) = \min_{x \in V \setminus R^{(k)}} \ell^{(k)}(x) \leq \ell^{(k+1)}(y).$$

Hence, (a) holds also after ③ and ④ since $R^{(k+1)} = R^{(k)} \cup \{u\}$.

In order to show that (b) is true after ④, we just have to consider the vertex u. Since (c) holds for k, it suffices to show that there exists no path π in G from s to u with a vertex $y \in V \setminus R^{(k+1)}$ and $\omega(\pi) < \ell^{(k)}(u) = \ell^{(k+1)}(u)$.

We assume there is such a path π, for example

$$\pi = \underbrace{s, v_1, \ldots, v_r}_{\in R^{(k)}}, \underbrace{y}_{\notin R^{(k)}}, v_{r+1}, \ldots, v_{r+m}, u.$$

Since (c) holds for k, we have $\ell^{(k)}(y) = \omega(s, v_1, \ldots, v_r, y)$, and it follows in view of the nonnegativity of the weights that

$$\omega(s, v_1, \ldots, v_r, y) \leq \omega(\pi) < \ell^{(k)}(u).$$

This means $\ell^{(k)}(y) < \ell^{(k)}(u)$ which contradicts $\ell^{(k)}(u) = \min_{x \in V \setminus R^{(k)}} \ell^{(k)}(x)$.

We verify next that also (c) is true after ③ and ④. If for some $v \in V \setminus R^{(k+1)}$ in ④

$$p^{(k+1)}(v) := u, \qquad \ell^{(k+1)}(v) := \ell^{(k+1)}(u) + \omega(u, v)$$

is set, then there must exist a path from s to v in the subgraph $G'(v)$ of G, which is generated by the vertices $R^{(k+1)} \cup \{v\}$. Its length is $\ell^{(k+1)}(u) + \omega(u, v)$ and its last edge is (u, v).

We assume that there exists a $v \in V \setminus R^{(k+1)}$ and a path π from s to v in $G'(v)$ with $\omega(\pi) < \ell^{(k+1)}(v)$. The vertex u must be contained in π, since only u has been added to $R^{(k+1)}$ and otherwise a contradiction to (c) would appear before the execution of ③ and ④ (as $\ell^{(k+1)}(v)$ is at most reduced compared to $\ell^{(k)}(v)$).

Let x denote the predecessor of v in π. Due to $x \in R^{(k+1)}$, (a) implies

$$\ell^{(k+1)}(x) \leq \ell^{(k+1)}(u)$$

while ④ implies

$$\ell^{(k+1)}(v) \leq \ell^{(k+1)}(x) + \omega(x, v) \leq \ell^{(k+1)}(u) + \omega(x, v) \leq \omega(\pi).$$

Here, the last inequality holds since π contains the vertex u and the edge (x, v). The inequality

$$\ell^{(k+1)}(v) \leq \omega(\pi)$$

contradicts however our assumption.

The propositions (a)–(c) hold therefore at any iteration index k. In particular, (b) holds when the algorithm terminates, which means that the algorithm works correctly.

Finally, the complexity estimate is obvious: The algorithm executes $n = |V|$ iterations, each of which has the complexity $\mathcal{O}(n)$. □

In case of graphs with *negative* weights but without negative cycles, one has to use the follwoing more costly algorithm. It is the most efficient algorithm known for this problem.

Algorithm 4.11 (Moore-Bellman-Ford)

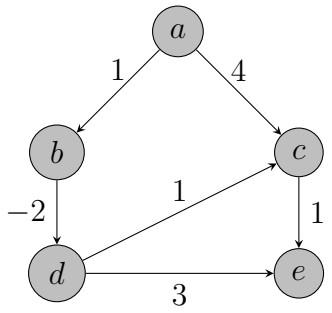
input: A weighted graph $G = (V, E, \omega)$ without negative cycles and an initial vertex $s \in V$.

output: Shorted paths from s to all $v \in V$ together with the length $\ell(v)$. Precisely, $\ell(v)$ is the length of the shortest s-v-path, which consists of the shortest s-$p(v)$-path and the edge $\big(p(v), v\big)$. For $v \notin \mathrm{post}^\star(s)$, $\ell(v) = \infty$ and $p(v)$ is undefined.

① set $\ell(s) := 0$ and $\ell(v) := \infty$ for all $v \in V \setminus \{s\}$

② for all $k = 1, 2, \ldots, n-1$:
  for each edge $(u, v) \in E$:
    if $\ell(v) > \ell(u) + \omega(u, v)$, then
      set $\ell(v) := \ell(u) + \omega(u, v)$ and $p(v) := u$

Example 4.12 Consider the weighted graph $G = (V, E, \omega)$ given by



Starting from the initial vertex a, the algorithm performs the following steps in the first iteration when we assume the edges are arranged in lexicographic order:

Edge	a	b	c	d	e	$p(b)$	$p(c)$	$p(d)$	$p(e)$
(a,b)	0	1	∞	∞	∞	a	—	—	—
(a,c)	0	1	4	∞	∞	a	a	—	—
(b,d)	0	1	4	-1	∞	a	a	b	—
(c,e)	0	1	4	-1	5	a	a	b	c
(d,c)	0	1	0	-1	5	a	d	b	c
(d,e)	0	1	0	-1	2	a	d	b	d

In the second iteration, only the length of the path to e is updated:

Edge	a	b	c	d	e	$p(b)$	$p(c)$	$p(d)$	$p(e)$
(a,b)	0	1	0	-1	2	a	d	b	d
(a,c)	0	1	0	-1	2	a	d	b	d
(b,d)	0	1	0	-1	2	a	d	b	d
(c,e)	0	1	0	-1	1	a	d	b	c
(d,c)	0	1	0	-1	1	a	d	b	c
(d,e)	0	1	0	-1	1	a	d	b	c

In the third iteration, nothing changes and, hence, the algorithm has converged. △

Theorem 4.13 (Moore-Bellman-Ford) The Moore-Bellman-Ford algorithm works correct. Its runtime is $\mathcal{O}(m \cdot n)$ with $m = |E|$ and $n = |V| \leq m$.

Proof. The complexity estimate is obvious.
At each point of time of the algorithm, let denote

$$R := \{v \in V : \ell(v) < \infty\},$$
$$F := \{(u,v) \in E : u = p(v)\}.$$

Then, we show:

(a) $\ell(v) \geq \ell(u) + \omega(u,v)$ for all $(u,v) \in F$,

(b) the graph (R,F) is acyclic,

(c) the graph (R,F) is a directed tree with initial vertex s, i.e., each vertex $v \in R$ is reachable from s by exactly one path.

When $p(v) := u$ is set in ②, then it just holds

$$\ell(v) = \ell(u) + \omega(u, v).$$

Since $\ell(u)$ is at most reduced, we deduce (a).
In order to verify (b), we assume that a cycle

$$\pi = v_0, v_1, \ldots, v_{r-1}, v_r, \quad v_0 = v_r$$

is generated at some time by setting $p(v_r) := v_{r-1}$. Then, however, it must be true before that

$$\ell(v_0) = \ell(v_r) > \ell(v_{r-1}) + \omega(v_{r-1}, v_r)$$

and according to (a)

$$\ell(v_i) \geq \ell(v_{i-1}) + \omega(v_{i-1}, v_i), \quad i = 1, 2, \ldots, r-2.$$

Summing up yields

$$\sum_{i=1}^{r} \omega(v_{i-1}, v_i) = \left(\sum_{i=1}^{r-1} \underbrace{\omega(v_{i-1}, v_i)}_{\leq \ell(v_i) - \ell(v_{i-1})} \right) + \underbrace{\omega(v_{r-1}, v_r)}_{< \ell(v_r) - \ell(v_{r-1})} < \sum_{i=1}^{r} \big(\ell(v_i) - \ell(v_{i-1})\big) = 0,$$

i.e., the cycle is of negative length, which contradicts the theorem's supposition.
Finally, $x \in R \setminus \{s\}$ implies also $p(x) \in R$, which is (c).
According to (a)–(c), $\ell(y)$ is at any point of time at least the length of the (unique) s-y-path in the graph (R, F). We proof next that $\ell(y)$ is after k iterations also at most the length of a shortest s-y-path in G with at most k edges. As the proposition is obvious for $k = 1$, we assume that it is true also after iteration $k - 1$. Consider a shortest s-y-path

$$\pi = s, v_1, \ldots, v_r, x, y, \quad r \leq k - 2$$

in G with at most k edges. Thus, $\pi' = s, v_1, \ldots, v_r, x$ is a shortest s-x-path with at most $k-1$ edges. This implies in view of the induction hypothesis that $\ell(x) \leq \omega(\pi')$ and hence

$$\ell(y) \leq \ell(x) + \omega(x, y) \leq \omega(\pi') + \omega(x, y) \leq \omega(\pi).$$

We conclude the correctness of the algorithm since there is no path with more than $n-1$ edges. □

We finally consider an algorithm for solving the all pair shortest path problem. Without loss of generality, let the vertices be indicated by the numbers $1, 2, \ldots, n$.

Algorithm 4.14 (Floyd-Warshall)

input: A weighted graph $G = (V = \{1, 2, \ldots, n\}, E, \omega)$ without negative cycles.
output: Matrices $[\ell_{i,j}]_{1 \leq i,j \leq n}$ and $[p_{i,j}]_{1 \leq i,j \leq n}$ which contain the length $\ell_{i,j}$ and last edge $(p_{i,j}, j)$ of the shortest path from i to j, if such a path exists.

① set $\ell_{i,j} := \omega(i, j)$ for all $(i, j) \in E$
set $\ell_{i,j} := \infty$ for all $(i, j) \in (V \times V) \setminus E$ with $i \neq j$
set $\ell_{i,i} := 0$ for all i
set $p_{i,j} := i$ for all $i, j \in V$

② for all $j = 1, 2, \ldots, n$:
  for all $i = 1, 2, \ldots, n$ with $i \neq j$:
    for all $k = 1, 2, \ldots, n$ with $k \neq j$:
    if $\ell_{i,k} > \ell_{i,j} + \ell_{j,k}$, then
      set $\ell_{i,k} := \ell_{i,j} + \ell_{j,k}$ and $p_{i,k} := p_{j,k}$

Theorem 4.15 (Floyd-Warshall) The algorithm of Floyd und Warshall works correct. Its runtime is $\mathcal{O}(n^3)$ with $n = |V|$.

Proof. The statement concerning the runtime is clear. We again add the iteration index j as suffix to all variables and show: after j_0 outer iterations, $\ell_{i,k}^{(j_0)}$ is the length of a shortest i-k-path, consisting only of interior vertices $v \in \{1, 2, \ldots, j_0\}$ and with last edge $(p_{i,k}^{(j_0)}, k)$. This proposition is verified by induction over j_0.

The proposition holds true for $j_0 = 0$ in accordance with ①. Moreover, for $j_0 = n$, it implies the correctness of the algorithm. We assume that the proposition is true for $j_0 \in \{0, 1, \ldots, n-1\}$, i.e., $\ell_{i,k}^{(j_0)}$ contains for all $i, k \in V$ a shortest path i-k-path consisting only of interior vertices $v \in \{1, 2, \ldots, j_0\}$.

In the (j_0+1)-th iteration, $\ell_{i,k}^{(j_0)}$ is replaced by $\ell_{i,j_0+1}^{(j_0)} + \ell_{j_0+1,k}^{(j_0)}$ provided this value is smaller. It remains thus to show that then the respective paths

$$\pi_1 = i, u_1, \ldots, u_r, j_0 + 1,$$
$$\pi_2 = j_0 + 1, v_1, \ldots, v_s, k$$

have no common interior vertices.

Assume that both paths have the vertex $u_p = v_q$ in common. Then,

$$\pi = i, u_1, \ldots, u_p, v_{q+1}, \ldots, v_s, k$$

is a path from i to k, consisting only of the vertices $v \in \{1, 2, \ldots, j_0\}$. Due to

$$\omega(u_p, u_{p+1}, \ldots, u_r, j_0 + 1, v_1, \ldots, v_q) \geq 0,$$

there follows

$$\ell_{i,k}^{(j_0)} \leq \omega(\pi) \leq \omega(\pi_1 \cup \pi_2) \leq \ell_{i,j_0+1}^{(j_0)} + \ell_{j_0+1,k}^{(j_0)},$$

which contradicts $\ell_{i,k}^{(j_0)} > \ell_{i,j_0+1}^{(j_0)} + \ell_{j_0+1,k}^{(j_0)}$. □